

Lec 21/22: Network layer numericals

Computer Networks (CSE 232)

Rinku Shah



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Solutions to classful addressing problems

Classless addressing

1. Subnetting
2. Supernetting

Problem 1

An organisation requires 30 different physical networks. Each network has about 2000 hosts.

- a. Which class should we choose for each network? What is the address utilization efficiency?
- b. Can a single class B network, say 190.32.0.0/16, suffice for 30 networks? If yes, what network addresses are assigned to 30 physical networks?
- c. Using the efficient solution, what is the network address for 190.32.11.5?
- d. How does this solve the address scalability problem?

Problem 2

An organisation has a physical network with 4000 hosts.

- a. To satisfy the required number of hosts, which class should we choose for each network? What is the address utilization efficiency?
- b. Can we assign multiple class C network addresses? If yes, what are the network addresses with minimal wastage of IP address if we start from 222.7.16.0/24?
- c. What is the problem with solution stated in (b)?
- d. How do we solve the problem in (b)?

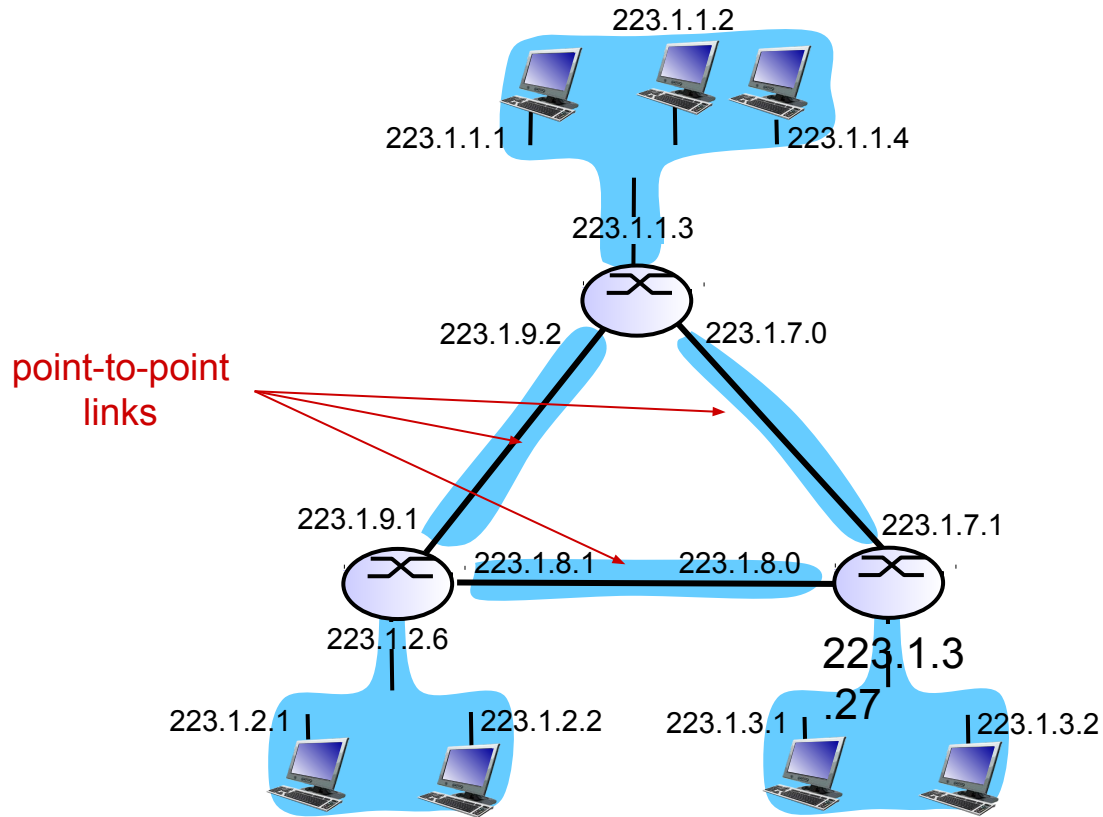
Problem 3

Suppose IIITD has assigned CSE department 10.129/16 prefix. We want to create 4 contiguous subnets for 4 labs that have 200, 100, 50, 50 computers respectively. What should be the network addresses for each lab so that there is minimal wastage of IP addresses?

Problem 4

Suppose IIITD has assigned CSE department 10.129/16 prefix. We want to create 4 contiguous subnets for 4 labs that have 400, 200, 100, 50 computers respectively. What should be the network addresses for each lab so that there is minimal wastage of IP addresses?

How many subnets do we have in this figure?



Destination-based forwarding

<i>forwarding table</i>	
Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through	2

Q: but what happens if ranges don't divide up so nicely?

Longest prefix matching

longest prefix matching

when looking for forwarding table entry for given destination address, use *longest* address prefix that matches destination address.

Destination Address Range	Link interface
11001000 00010111 00010*** *****	0
11001000 00010111 00011000 *****	1
11001000 00010111 00011*** *****	2

examples. otherwise

DA: 11001000 00010111 00010110 10100001

DA: 11001000 00010111 00011000 10101010

which interface?

which interface?

Problem 5

The routing table of a router is shown below:

Destination	Subnet mask	Interface
11.0.0.0/8	255.0.0.0	Eth0
11.129.0.0/16	255.255.0.0	Eth1
11.129.0.0/24	255.255.255.0	Eth2
11.129.1.0/25	255.255.255.128	Eth3
11.129.1.128/26	255.255.255.192	Eth4
11.129.1.192/26	255.255.255.192	Eth5
default		Eth6

$$11.129.1.140 \wedge 255.0.0.0 = 11.0.0.0$$

$$11.129.1.140 \wedge 255.255.0.0 = 11.129.0.0$$

$$11.129.1.140 \wedge 255.255.255.0 = 11.129.1.0$$

$$11.129.1.140 \wedge 255.255.255.128 = 11.129.1.128$$

$$11.129.1.140 \wedge 255.255.255.192 = 11.129.1.128$$

$$11.129.1.140 \wedge 255.255.255.192 = 11.129.1.128$$

On which interface will the router forward packets addressed to destination 11.129.1.140?

Longest prefix matching

- longest prefix matching: often performed using ternary content addressable memories (TCAMs)
 - *content addressable*: present address to TCAM: retrieve address in one clock cycle, regardless of table size
 - Cisco Catalyst: can up ~1M routing table entries in TCAM

Problem 6

The routing table of a router is shown below:

Destination	Subnet mask	Interface
128.75.43.0	255.255.255.0	Eth0
128.75.43.0	255.255.255.128	Eth1
192.12.17.5	255.255.255.255	Eth3
default		Eth2

On which interface will the router forward packets addressed to destinations 128.75.43.16 and 192.12.17.10 respectively?

How are IP addresses allocated?

1. Get an IP address from an ISP
 - a. For example,
 - i. suppose an ISP assigns 150.5.0.0/24 to the customer network
 - ii. ISP announces a network with bigger prefix, 150.5.0.0/16
 - iii. When traffic with /24 prefix reaches the ISP, the traffic is routed to the customer network
 - b. Problem: If you change the ISP, you have to renumber your hosts
2. Apply and get your own IP address prefix from Internet registries
 - a. Announce this prefix via the ISPs
3. Once you have a prefix, how do you assign IP addresses?
 - a. Announce statically or dynamically using DHCP

Our solution to problem 3 had following 4 subnets: 10.129.0.0/24, 10.129.1.0/25, 10.129.1.128/26, 10.129.1.192/26

For instance, assume these are public IP addresses then the ISP will announce a network with prefix 10.129.0.0/16 and route packets with /24, /25, /26 prefixes to the organisation.

Typical router architecture

